Supplementary methods

Cross-study synthetic-guided transcriptomics in spaceflown mice

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> **Frozen analysis supplement.** This document records exact data roles, architecture, evaluation gates, statistical safeguards, output locations, and rebuild commands. It does not rerun training.

S1. Reproducibility contract

The manuscript builder consumes completed outputs and fails when a required file or a key expected result is missing. It does not:

- query the OSDR API;
- preprocess expression;
- train or fine-tune a neural network;
- sample a new synthetic cohort;
- rerun feature selection;
- recalculate random-effects statistics from raw profiles.

The exact frozen inputs and SHA-256 hashes are in `source_data/frozen_input_manifest.tsv`. Figure hashes are in `source_data/figure_build_manifest.tsv`.

S2. OSDR discovery and expression ingestion

OSDR data were obtained through the Biological Data API documented at https://visualization.osdr.nasa.gov/biodata/api/. The repository implementation and endpoint notes are in `docs/osdr_api.md`.

Eligibility required:

- `Mus musculus`;
- transcription profiling by bulk RNA sequencing;
- a resolvable spaceflight/flight or ground-control label;
- processed RSEM expected-count output;
- sample-level tissue or material metadata sufficient for canonicalization.

No raw combined OSDR HDF5 file was used. The API audit outputs are:

```
outputs/generative_benchmark/data_audit/osdr/osdr_canonical_metadata.tsv
outputs/generative_benchmark/data_audit/osdr/osdr_inventory_summary.json
outputs/generative_benchmark/data_audit/osdr/osdr_tissue_alias_audit.tsv
outputs/generative_benchmark/data_audit/osdr/osdr_tissue_inventory.tsv
```

The API returned 1,631 profile rows. Twenty-one technical replicates were aggregated, leaving 1,610 biological profiles, 835 flight and 775 ground control, from 75 accessions and 24 canonical material classes.

S3. ARCHS4 cohort audit

The source file was `assets/archs4/mouse_gene_v2.5.h5`, containing 997,515 profiles and 53,511 genes. The complete audit files are:

```
outputs/generative_benchmark/data_audit/archs4/archs4_full_profile_audit.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_control_only_balanced.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_healthy_preferred_balanced.tsv.gz
outputs/generative_benchmark/data_audit/archs4/archs4_broad_balanced.tsv.gz
```

The three eligible reference cohorts contained:

Cohort	Profiles	GEO series	Intended use
Control only	23,614	3,213	Conservative sensitivity cohort
Healthy preferred	62,299	5,307	Primary pretraining source
Broad	134,250	15,111	Diversity sensitivity cohort

Before reference selection, nine GEO series linked to eligible OSDR accessions were excluded by accession metadata. This removed 108 otherwise eligible ARCHS4 profiles, including all 53 selected profiles from GSE152382 (OSD-457). The paper-parity run then selected 17,244 healthy-preferred profiles across 20 tissue classes. Complete GEO-series grouping assigned 10,150 profiles to training, 2,466 to validation, and 4,628 to test, with no series shared across splits.

The exclusion list and overlap audit are:

```
data/diffusion/osdr_archs4_overlap_exclusions.tsv
docs/diffusion_leak_free_confirmation.md
```

S4. Landmark panel and normalization

Full-transcriptome TPM was calculated with `data/reference/gencode_vM39_mouse_gene_lengths.tsv`. Landmark selection occurred after TPM calculation. The deterministic 974-gene panel and the human-to-mouse mapping audit are:

```
data/diffusion/l974_mouse_paper_parity.tsv
data/diffusion/l1000_human_to_mouse_ensembl.tsv
```

Training-set MaxAbs scaling was applied after landmark selection. The ARCHS4 prepared matrix is:

```
outputs/generative_benchmark/data/lacan_diffusion/archs4_mouse_paper_parity_osdr_disjoint_l974.h5
```

The OSDR factorized matrix and profile metadata are:

```
outputs/generative_benchmark/data/lacan_diffusion/
osdr_factorized_within_study_replicated_validation_l974.h5
outputs/generative_benchmark/data/lacan_diffusion/
osdr_factorized_within_study_replicated_validation_l974.samples.tsv.gz
```

S5. ARCHS4 DDIM configuration

The full configuration is `configs/rna_diffusion/archs4_mouse_paper_parity_osdr_disjoint.yaml`. The neural architecture and optimizer settings match the paper-parity implementation; only the reference exclusion and grouped split contract changed.

Component	Value
Input genes	974
Hidden layers	8,192; 8,192
Parameters	227,109,786
Dropout	0.1
Diffusion steps	1,000
Beta schedule	Quadratic, 0.0001 to 0.02
Objective	Summed noise MSE
Optimizer	Adam
Learning rate	0.0004783833151836702
Scheduler	OneCycle
Batch size	2,048
Epochs / optimizer steps	15,000 / 75,000
AMP	Enabled
EMA	0.999
Device	NVIDIA A100-SXM4-40GB
Runtime	6,083 seconds
Peak allocated GPU memory	5.93 GB

Run directory:

```
outputs/generative_benchmark/runs/lacan_diffusion/
archs4_mouse_paper_parity_osdr_disjoint_seed1234/
```

S6. Factorized OSDR adaptation

The accepted configuration is `configs/rna_diffusion/osdr_factorized_study_lora512_correlation_refine_osdr_disjoint.yaml`. It uses the OSDR-disjoint ARCHS4 backbone described in Section S5. A base adapter was trained for 12,000 domain and 4,000 condition steps, followed by the 4,000-domain-step and 1,000-condition-step correlation-refinement stage reported here. The all-tissue and muscle-group development screens were then regenerated from this adapter. Section S10 describes the separate lung/thymus accession-held-out experiment.

Component	Value
Backbone	Completed ARCHS4 paper-parity DDIM
Conditioning	Tissue, FLT/GC, accession, material type
Domain adapter	LoRA rank 512, alpha 512
Domain stage	4,000 steps, LR 0.00002
Condition stage	1,000 steps, LR 0.00002
Batch size	512
Condition dropout	0.15
Correlation regularization	Weight 10, 256 genes, timestep <= 200
Sampling	100 DDIM steps for evaluation

The data split contained 781 training, 536 validation, and 293 locked-test profiles. Every test accession was represented in training. Results therefore measure within-study interpolation.

The accepted calibrator used:

1. train-only global alignment;
2. hierarchically shrunk accession and tissue means;
3. positive missing-covariance residual noise;
4. no condition-specific fit of calibrator means or covariance;
5. explicit clipping at zero for accepted downstream expression.

Run directory:

```
outputs/generative_benchmark/runs/lacan_diffusion/
osdr_factorized_study_lora512_correlation_refine_osdr_disjoint_seed2020/
```

S7. Distribution and condition gates

Four generation seeds, 5020-5023, were declared before the locked test was opened. Every metric was gated independently.

Metric	Gate
Gene-correlation agreement	Paper target >= 0.98; finite-sample real-bootstrap floor also reported
Precision	>= 0.95
Recall	>= 0.85
F1	>= 0.90
Adversarial accuracy	0.40 to 0.60
FD / real-split P95	<= 1.0
Pooled effect recovery	Correlation >= 0.30 and direction >= 0.55
Muscle accession recovery	Correlation >= 0.30 and direction >= 0.55
Memorization	Generated fraction below train LOO P01 <= 0.05

The exact repeat rows are in `source_data/table_s2_locked_ddim_repeats.tsv`. On the locked within-study test, mean gene-correlation agreement was 0.9744, precision 0.9974, recall 0.9957, F1 0.9966, adversarial accuracy 0.4753, and FD/real-split-P95 ratio 0.0740. All four repeats passed the finite-sample correlation floor of 0.9497, although the mean remained below the paper target of 0.98. Pooled FLT/GC effect recovery passed in three of four repeats and muscle accession-effect recovery passed in all four. The preceding validation screen missed its stricter sample-specific correlation floor and muscle accession-effect gate; the broad biological screens therefore remain developmental rather than study-held-out confirmation.

The term "adversarial accuracy" refers to an external nearest-neighbor real-versus-synthetic classifier, not the WGAN training critic. A result near 0.5 indicates that this external discriminator cannot reliably separate the two cohorts.

S8. WGAN-GP and GeneJEPA screens

The WGAN used the Viñas et al. topology: 64-dimensional noise, two 256-unit generator and critic layers, five critic updates, gradient-penalty weight 10, RMSProp learning rate 0.0005, and batch size 32. The strongest study-conditioned validation result had external adversarial accuracy 0.6362. Because no calibration variant jointly fixed adversarial accuracy and retained the correlation floor, and because accession-aware FLT/GC recovery failed, its locked test remained unopened.

```
outputs/generative_benchmark/runs/vinas_wgan_gp/osdr_matched_study_conditioned_seed2020/
```

The exact-architecture GeneJEPA duration screen used 4,096 genes and 43,744 replacement-sampled training exposures. It reached 0.703 held-out tissue balanced accuracy versus 0.839 from expression. It is representation-only and has no expression decoder.

```
outputs/generative_benchmark/runs/genejepa/
matrix_phase_0_genejepa_exact_mouse_one_epoch_f2e01cf1f130d5cb/
```

S9. Generated-feature workflow

Five arms were compared:

1. `real_only`;
2. `generated_only`;
3. `real_plus_generated`, equal total real and synthetic weight;
4. `guided_real_only`, real classifier with real/synthetic consensus ranking;
5. `guided_low_weight`, real plus recentered synthetic profiles at 0.05 total synthetic weight.

Splits were nested within accession-by-condition strata. The inner loop selected feature count, regularization, and rank method. The outer loop measured balanced accuracy, AUROC, and average precision separately. No composite score was used.

Stable genes had selection frequency at least 0.50 and coefficient-sign agreement at least 0.75. Generated-supported status did not constitute biological evidence. Real random-effects and LOO tests were applied afterward.

For the BH-FDR synthetic-informed subset, `reinforced` denotes a `core_intersection` gene that was stable in both the real-only and selected synthetic-guided arms, had matching coefficient directions, and had a supporting real meta-effect. `Synthetic-promoted` denotes a `generated_supported` gene that was stable in the selected synthetic-guided arm but did not cross the real-only stability threshold, with a supporting real meta-effect. Thus `synthetic-promoted` is a feature-selection classification, not a claim that the gene was absent from real expression, statistically nonsignificant in real data, or biologically novel.

Primary workflow documentation:

```
docs/generated_feature_guidance_workflow.md
```

S10. Held-out study test

The confirmation protocol is frozen at:

```
outputs/generative_benchmark/analyses/generated_feature_guidance_confirmation_disjoint_v1/
protocol.md
```

Test accessions:

Tissue	Accession	Profiles	FLT	GC
Lung	OSD-900	20	10	10
Thymus	OSD-457	24	12	12

Both accessions were removed from all OSDR generator-adaptation roles. A post-analysis audit found that the original ARCHS4 reference included GEO series linked to OSDR, including GSE152382 from OSD-457. Three exact OSD-457 thymus profiles had been assigned to the original ARCHS4 training split. That overlap invalidated the original fully unseen wording.

For the final analysis, all nine OSDR-linked GEO series were excluded before reference selection, the 15,000-epoch ARCHS4 backbone was trained from scratch, and the 5,000-epoch OSDR adaptation and fixed feature-guidance protocol were rerun. OSD-464 lung and OSD-244 thymus remained fixed validation studies. Because the original outcomes had already been observed before the final run, this is not a pristine prospective confirmation.

The deployed thymus classifier used real profiles only. Synthetic data changed feature ranking. The deployed lung classifier used a recentered synthetic view at 0.05 total sample weight during policy evaluation, but failed the prespecified inner gate; the deployed lung result therefore reverted to the real-only baseline.

Tissue	Baseline BA	Guided BA	Baseline AUROC	Guided AUROC	Baseline AP	Guided AP
Lung, OSD-900	0.400	0.350	0.450	0.470	0.523	0.578
Thymus, OSD-457	0.500	0.833	0.840	0.979	0.876	0.983

Genotype assignment was audited after the primary result:

Tissue	Stratum	Profiles	FLT	GC
Lung	KO	10	5	5
Lung	WT	10	5	5
Thymus	Nrf2KO	12	6	6
Thymus	WT	12	6	6

Relationship between evidence tiers

Tier 1 asks whether a frozen synthetic-guided policy transfers to an OSDR accession excluded from adaptation and feature-policy development after OSDR-linked GEO series are also removed from ARCHS4. It remains retrospective because an earlier run exposed the outcomes. Tier 2 asks which BH-significant real effects also cross repeated synthetic-informed selection thresholds within the development domain. Tier 3 is the complete real-data random-effects BH-FDR screen, regardless of feature-selection status. These tiers answer different questions and are not alternative statistical filters on one gene list.

All eight Tier 1 thymus genes were FLT-lower in both OSD-457 genotype strata and had FLT-lower cross-study meta-effects with BH FDR < 0.05. Their separate Tier 2 labels were:

Gene	OSD-457 result	Tier 2 selection label	Cross-study BH FDR
`Birc5`	FLT lower in WT and Nrf2KO	Synthetic-promoted	`1.26e-7`
`Cdk1`	FLT lower in WT and Nrf2KO	Synthetic-promoted	`4.83e-7`
`Ccnb2`	FLT lower in WT and Nrf2KO	Synthetic-promoted	`2.38e-4`
`Nusap1`	FLT lower in WT and Nrf2KO	Synthetic-promoted	`1.16e-9`
`Ccnb1`	FLT lower in WT and Nrf2KO	Not selection-stable	`3.40e-10`
`Gmnn`	FLT lower in WT and Nrf2KO	Reinforced	`0.0227`
`Ccne2`	FLT lower in WT and Nrf2KO	Synthetic-promoted	`0.00172`
`Ube2c`	FLT lower in WT and Nrf2KO	Reinforced	`0.0102`

This mapping is frozen in Supplementary Table S19. The Tier 1 conclusion is that synthetic guidance assembled a coherent, transferable cell-cycle panel from real-supported genes. The Tier 2 labels describe thresholded selection behavior and should not be interpreted as eight independent novelty claims.

The all-tissue and muscle-group development screens were rerun with the same backbone. Synthetic attribution was retained only when the selected generated arm was nonworse than real-only balanced accuracy, AUROC, and average precision under the frozen selection rule. Random-effects BH-FDR values were calculated only from real OSDR profiles.

S11. Random-effects reporting and LOO sensitivity

For gene (g) in accession (a), the real flight effect was:

```
delta[g,a] = mean(real expression[g] | FLT,a)
           - mean(real expression[g] | GC,a)
```

Accession effects were combined with a random-effects model. Within each tissue, Benjamini-Hochberg (BH) adjustment was applied to the 974 gene-level meta-analysis P values. BH sorts the P values and controls the expected proportion of false discoveries among the rejected hypotheses. The primary inclusion rule was BH FDR < 0.05; synthetic selection, study-direction agreement, heterogeneity, and LOO stability were not additional significance gates.

After BH correction, each association was annotated as:

- reinforced by stable real-only and synthetic-guided selection;
- synthetic-promoted;
- selected by the real-only arm;
- synthetic-selected without real-direction support; or
- not stably selected by either feature-selection arm.

The accession-direction fraction, random-effects variance (`tau2`), heterogeneity (`I2`), and exact study counts remain in the complete inventory. A gene with mixed study directions remains BH-significant when its pooled random-effects test passes, but the disagreement qualifies interpretation of the pooled effect.

The LOO analysis removed each accession, repeated the random-effects fit, and retained the maximum FDR and any sign reversal. LOO was treated as a sensitivity label rather than an inclusion requirement. A LOO-stable gene additionally required maximum LOO FDR < 0.05 and no LOO sign reversal. Failure of this label does not remove a gene from the primary BH-FDR table.

Generated profiles were never included in the random-effects model.

S12. Reactome analysis

The official mouse GMT is:

```
data/pathways/reactome_current_mouse_ensembl.gmt
```

It was generated from official `ReactomePathways.txt` and `Ensembl2Reactome_All_Levels.txt`, restricted to *Mus musculus*, `R-MMU-*` pathways, and `ENSMUSG*` genes.

Hypergeometric enrichment used the 974-gene landmark panel as background. Benjamini-Hochberg FDR was applied separately by tissue and selected-gene set. Reactome parent and child terms overlap. Counts of significant rows are therefore not counts of independent biological discoveries.

S13. Skeletal-muscle group analysis

The factorized DDIM and three frozen synthetic development views were reused. No additional neural network was trained for the muscle-group screen.

Group	Profiles	Accessions	FLT	GC
EDL	24	2	12	12
Gastrocnemius	25	3	10	15
Quadriceps	35	4	18	17
Soleus	41	3	22	19
Tibialis anterior	24	2	12	12

The full report is `docs/synthetic_skeletal_muscle_group_analysis.md`. Key frozen outputs are:

```
outputs/generative_benchmark/analyses/
within_study_generated_feature_stability_muscle_groups_osdr_disjoint_v1/
```

The selected soleus arm was real plus generated expression. It improved mean balanced accuracy from 0.9250 to 0.9625, left AUROC at 0.9800, and increased average precision from 0.9804 to 0.9865. Five BH-FDR genes were reinforced by both stable real-only and selected-arm feature ranking and had the same real effect direction in all three accessions: FLT-lower `Bdh1`, `Ech1`, `Bnip3`, and `Decr1`, and FLT-higher `Tpm1`. `Bdh1`, `Ech1`, `Bnip3`, and `Tpm1` also passed the LOO sensitivity criterion; `Decr1` did not. The screen did not promote `Mef2c` or `Pxmp2`.

Gastrocnemius selected a guided low-weight arm and promoted `Fhl2` and `Nfkbia`. Tibialis anterior selected real plus generated expression and reinforced `Cdkn1a`, `St3gal5`, and `Bnip3` while promoting `Cebpd`. None of these gastrocnemius or tibialis genes passed the LOO FDR sensitivity rule. EDL and quadriceps retained real-only arms, so their BH-FDR genes are not attributed to synthetic guidance.

LOO here is a real-data meta-analysis sensitivity test. It does not remove the accession from the already completed generator adaptation. Soleus remains developmental until a new accession is excluded from adaptation and selection.

S14. Spleen `Igfbp3` follow-up

The all-tissue screen did not stably select `Igfbp3` (`ENSMUSG00000020427`) in either the real-only or selected synthetic-guided arm. It therefore does not demonstrate a synthetic contribution. The separate real-data follow-up remains a secondary biological result: flight-minus-ground effects were positive in OSD-164, OSD-246, OSD-288, OSD-420, OSD-457, and OSD-506; random-effects FDR was `1.76e-09`; and the maximum FDR across the six leave-one-accession-out analyses was `0.00385`.

A separate TPM calculation from the API-derived full-transcriptome count matrix found flight/ground-control mean ratios of 1.09 to 1.63 across the six studies. These ratios are descriptive; the accession-aware random-effects analysis is the formal statistical result.

Normal spleen source localization was performed after gene discovery:

- GSE156162 sorted-cell data placed the highest baseline `Igfbp3` expression in white-pulp mesenchymal cells, followed by red-pulp mesenchymal cells.
- E-MTAB-7703 enriched stromal single-cell data localized expression to fibroblastic reticular, collagen-producing, and perivascular populations.
- Flight splenocyte, PBMC, and marrow single-cell datasets lacked sufficient signal to test those populations because their preparation excludes or strongly depletes nonhematopoietic spleen stroma.

These source-localization analyses are post hoc and do not constitute an independent flight replication. The defensible hypothesis is that whole-spleen `Igfbp3` elevation reflects altered stromal expression, altered stromal abundance or architecture, or both. It is presented as real-data context, not as a synthetic-guided discovery. The full audit and reproduction notes are in `docs/spleen_igfbp3_handoff.md`.

S15. Random-effects BH-FDR gene inventories

Supplementary Table S17 is the primary statistical inventory. It contains every real-data random-effects association with BH FDR < 0.05, without requiring synthetic selection, unanimous study direction, or LOO stability. Supplementary Table S18 reports the corresponding counts for every tested analysis unit, including tissues with zero BH-FDR genes.

The 459 retained tissue-gene results comprise 202 associations across 10 of 22 canonical tissues and 257 across all five anatomical muscle groups. This is an inventory size, not a count of independent discoveries: genes can recur across tissues, and pooled skeletal muscle overlaps the anatomical subgroup analyses. Of these results, 363 had the same effect direction in every represented accession and 96 did not. Direction agreement is reported as an annotation, not an exclusion rule.

Analysis unit	BH-FDR genes	FLT higher	FLT lower	Unanimous direction	Synthetic-promoted	Reinforced
Adrenal gland	22	1	21	21	1	1
Eye	8	4	4	8	0	1
Heart	4	2	2	4	0	0
Kidney	3	3	0	1	1	1
Liver	19	8	11	0	0	0
Retina	5	5	0	5	0	0
Skeletal muscle, pooled	26	12	14	5	4	8
Skin	3	2	1	1	1	0
Spleen	34	32	2	21	3	1
Thymus	78	37	41	57	13	3
EDL	136	14	122	130	0	0
Gastrocnemius	15	8	7	13	2	0
Quadriceps	29	25	4	22	0	0
Soleus	29	5	24	27	0	5
Tibialis anterior	48	42	6	48	1	3

Supplementary Table S16 is the 49-row synthetic-informed subset of the primary inventory: 26 synthetic-promoted and 23 reinforced tissue-gene results with supporting real effects. Synthetic attribution is suppressed when the candidate generated arm failed the three-metric eligibility gate. The table is retained to describe what the generator changed in feature ranking, but it is not the primary significance table. The full inventory additionally contains 34 real-only selected results, 370 BH-significant results not stably selected by either arm, and six synthetic-selected results without real-direction support.

The complete Tier 2 genes are shown below. `FLT higher` and `FLT lower` refer to the sign of the real random-effects meta-estimate. An asterisk marks a BH-significant pooled effect whose direction was not identical in every accession.

Synthetic-promoted Tier 2 genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	—	`Psmb8`
Kidney	`Inpp4b`*	—
Skeletal muscle, pooled	—	`Klhl21`*, `Mapkapk5`*, `Reep5`*, `Itgb5`*
Skin	`Plscr1`	—
Spleen	`Rai14`, `Myl9`*, `Ptprk`	—
Thymus	`Hsd17b11`*, `Etv1`	`Nusap1`, `Stmn1`, `Birc5`, `Cdk1`, `Top2a`, `Ccnb2`, `Aurka`, `Ccne2`, `Kif20a`, `Pcna`, `Ccnf`
Gastrocnemius	`Nfkbia`	`Fhl2`
Tibialis anterior	`Cebpd`	—

Reinforced Tier 2 genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	—	`Tspan4`
Eye	—	`Klhl21`
Kidney	`Slc37a4`	—
Skeletal muscle, pooled	`Sox4`, `Cebpd`*, `Sh3bp5`, `Prkcd`, `Arid5b`*, `Sesn1`*, `Tle1`*	`Bphl`*
Spleen	`Loxl1`	—
Thymus	`Snx7`	`Ube2c`, `Gmnn`
Soleus	`Tpm1`	`Bdh1`, `Ech1`, `Bnip3`, `Decr1`
Tibialis anterior	`Cdkn1a`, `St3gal5`, `Bnip3`	—

Heart, liver, retina, EDL, and quadriceps had Tier 3 BH-FDR genes but no Tier 2 synthetic-informed gene. Bone, bone marrow, brain, brown adipose tissue, cecum, cerebellum, colon, hippocampus, lung, mammary gland, optic nerve, and white adipose tissue had no Tier 3 BH-FDR gene in the 974-gene panel.

The pooled skeletal-muscle results remain auditable in Tables S17-S18 and are interpreted separately from anatomical groups because those groups have different responses. Liver has 19 real-data BH-FDR associations but selected a real-only arm. Skin has three real-data BH-FDR associations and one synthetic-promoted gene, `Plscr1`. Lung has no BH-FDR gene in the 974-gene panel.

S16. Supplementary figures

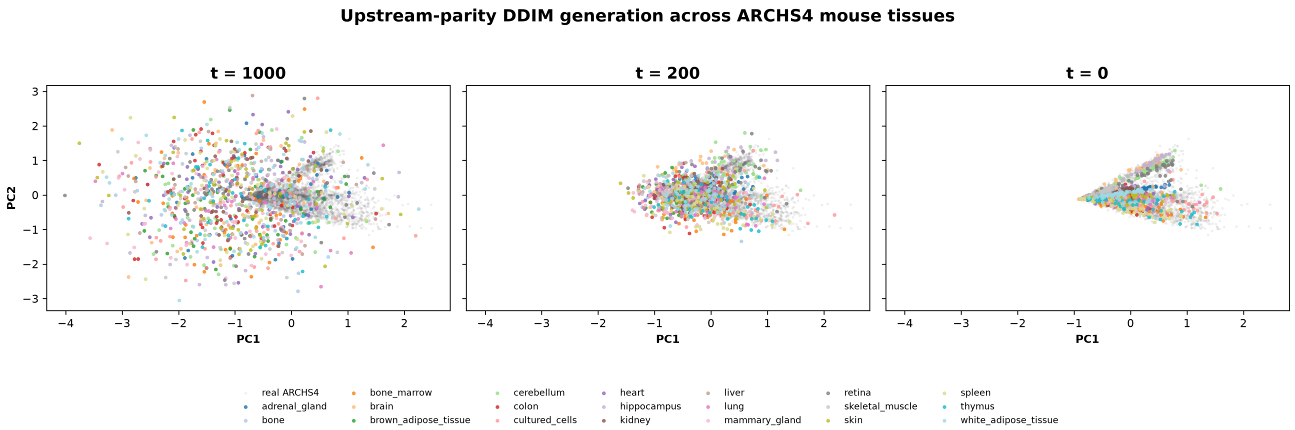

Figure S1. ARCHS4 DDIM denoising trajectory. The same generated profiles are shown at diffusion timesteps 1,000, 200, and 0 in a PCA space fitted to real ARCHS4 expression. Colors identify tissue classes. The two-dimensional view is descriptive; held-out tissue classification uses the full 974-gene representation.

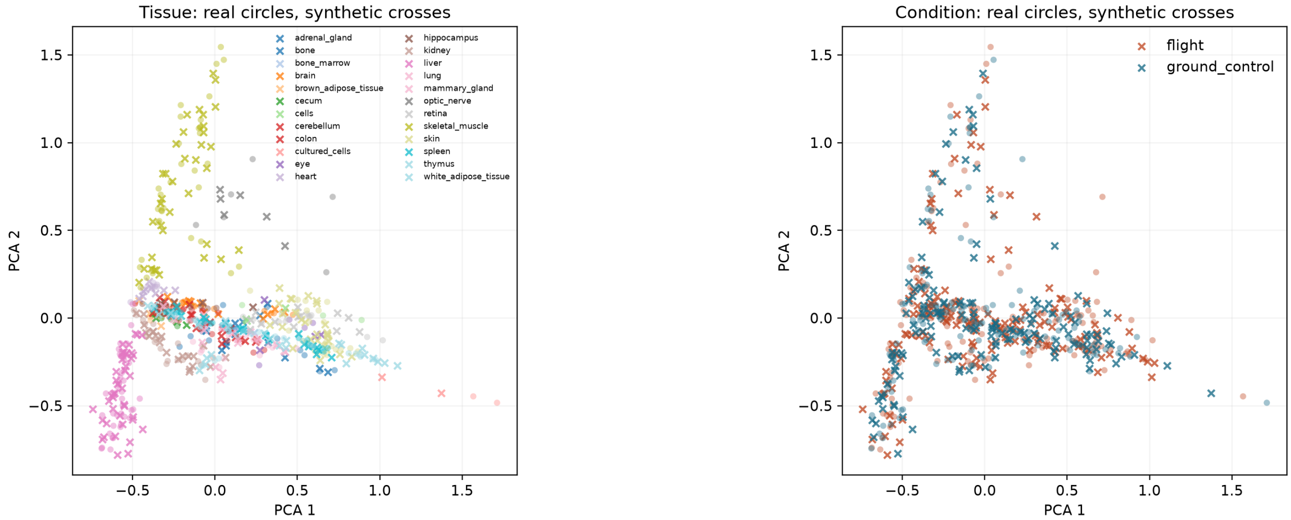

Figure S2. Real and generated profiles in the locked OSDR test. Seed 5020 is shown. Tissue and condition views are descriptive; formal fidelity and effect metrics use all declared seeds and higher-dimensional data.

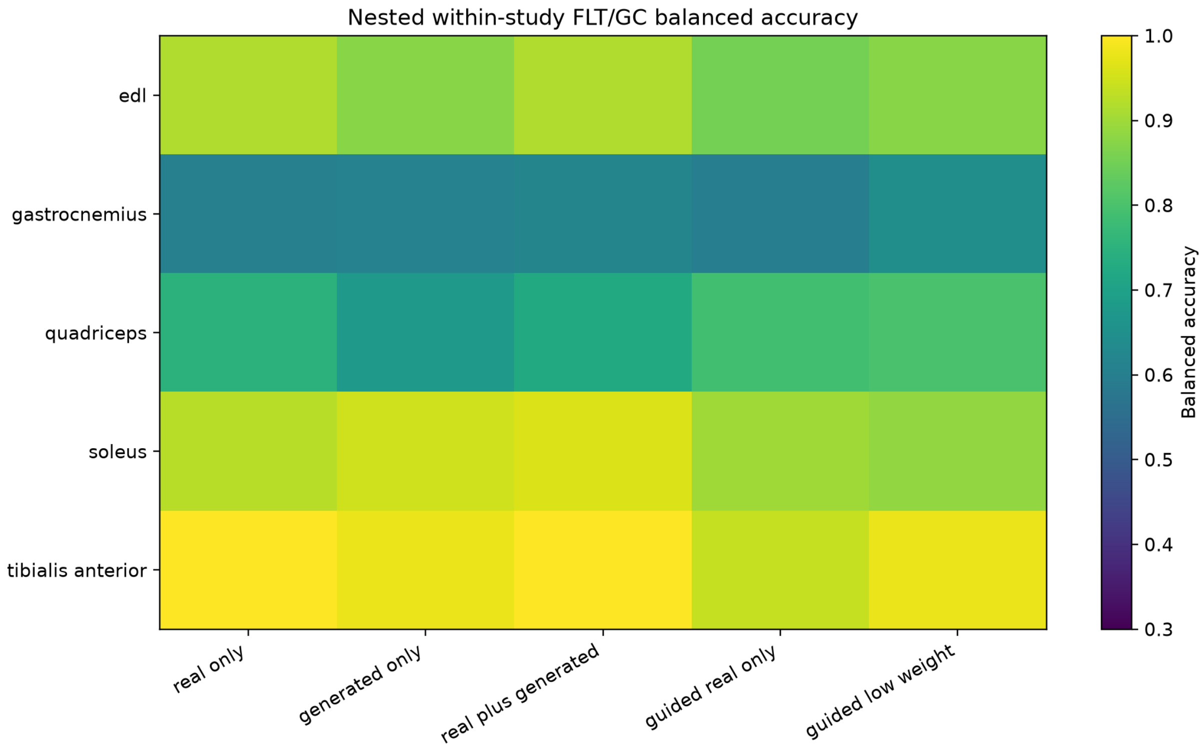

Figure S3. Repeated nested muscle-group balanced accuracy. Each row is a muscle group and each column is a downstream use of real or generated profiles. Arm selection also required nonworse AUROC and average precision.

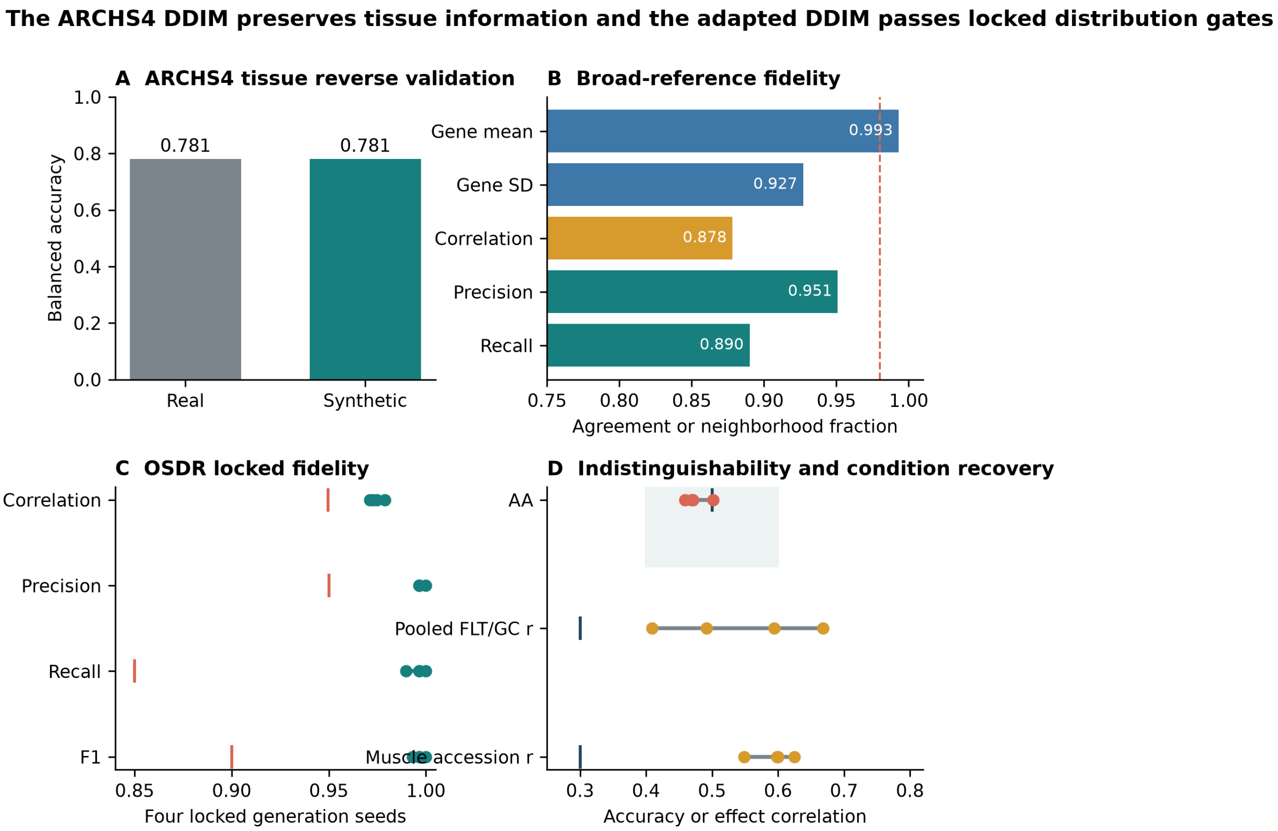

Figure S4. Generator validation. (A) Tissue balanced accuracy when a classifier was trained on held-out ARCHS4 real or synthetic profiles. (B) Broad-reference distribution metrics. The dashed line marks the strict correlation target. (C) Four OSDR locked-test generations; vertical marks show metric gates. (D) External adversarial accuracy and pooled or accession-aware flight-effect recovery. The shaded interval is the accepted adversarial-accuracy range.

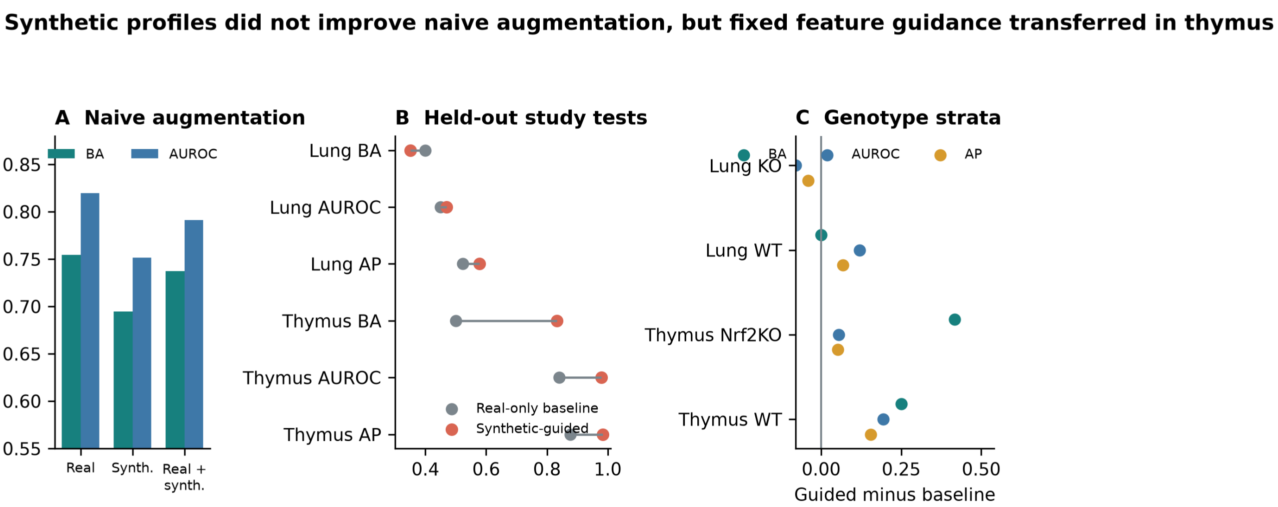

Figure S5. Downstream utility of generated expression. (A) Direct pooled augmentation on the locked real test. (B) Fixed synthetic-guided policies in OSDR-held-out lung and thymus accessions. (C) Guided-minus-baseline metric changes after post-hoc genotype stratification. Thymus improved uniformly; lung knockout AUROC declined.

S17. Source tables

- `table_1_data_inventory.tsv`
- `table_2_model_screen.tsv`
- `table_3_locked_ddim_metrics.tsv`

- `table_4_leakage_corrected_confirmation.tsv`
- `table_5_tissue_evidence.tsv`
- `table_s1_archs4_ddim_metrics.tsv`
- `table_s2_locked_ddim_repeats.tsv`
- `table_s3_naive_augmentation.tsv`
- `table_s4_confirmation_genotypes.tsv`
- `table_s5_thymus_core_genes.tsv`
- `table_s6_thymus_reactome.tsv`
- `table_s7_muscle_group_summary.tsv`
- `table_s8_soleus_genes.tsv`
- `table_s9_muscle_reactome.tsv`
- `table_s10_all_tissue_development_screen.tsv`
- `table_s11_spleen_igfbp3_accession_effects.tsv`
- `table_s12_spleen_igfbp3_random_effects.tsv`
- `table_s13_spleen_reference_expression.tsv`
- `table_s14_quadriceps_rbm6_accession_effects.tsv`
- `table_s15_quadriceps_rbm6_random_effects.tsv`
- `table_s16_ordinary_fdr_directional_genes.tsv`
- `table_s17_all_random_effects_bh_fdr_genes.tsv`
- `table_s18_bh_fdr_tissue_summary.tsv`
- `table_s19_thymus_evidence_level_mapping.tsv`

S18. Rebuild command

```
PYTHONPATH=src /home/exouser/miniforge3/envs/nasa-mouse/bin/python \
  -m nasa_mouse_rna_diffusion.build_synthetic_guided_paper
```

To regenerate source tables and figures without rendering PDFs:

```
PYTHONPATH=src /home/exouser/miniforge3/envs/nasa-mouse/bin/python \
  -m nasa_mouse_rna_diffusion.build_synthetic_guided_paper --skip-render
```